IPMAI / ACDP / INTEMAC Cycle Monitor SW – Quick Start Guide

Purpose:

This software package provides interactive tools for loading, analyzing, visualizing, and comparing measured industrial and control-system signals. It includes PID simulation analysis, cycle monitoring, feature extraction, raw data visualization, anomaly evaluation, and HTML documentation access.

Installation (Windows):

1. Install Python 3.11+.
2. Extract the ZIP package into a working directory.
3. Open Command Prompt in the extracted folder.
4. Install required packages:

```
pip install numpy scipy pandas matplotlib pyqtgraph pyqt6 h5py openpyxl
```

Folder Structure:

- /data – input XLS/XLSX/CSV measurement files
- /output – generated features, plots, HDF5 files, results
- /img – launcher images and graphics
- /docs or /html – generated HTML documentation
- run_0_*.py – main launcher scripts
- run_1_*.py – viewers and preprocessing modules
- run_2_*.py – feature extraction and analysis modules

Running the Software:

Start the main launcher from Command Prompt:

```
python run_0_main_launcher.py
```

The launcher allows starting individual viewers, processing pipelines, feature extraction tools, comparison viewers, and HTML documentation.

Typical Workflow:

1. Copy measurement files into /data.
2. Open Raw Signal Viewer and inspect signals.
3. Run preprocessing / cycle extraction.
4. Run feature extraction or anomaly analysis.
5. Open comparison viewers and exported plots/results.

Supported Data:

CSV, XLS, XLSX, and HDF5 files containing industrial process measurements, PID variables, actuator signals, encoder signals, vibration signals, or harmonic gearbox monitoring data.

Documentation:

Detailed documentation is available through the launcher or by opening index.html in the documentation folder.